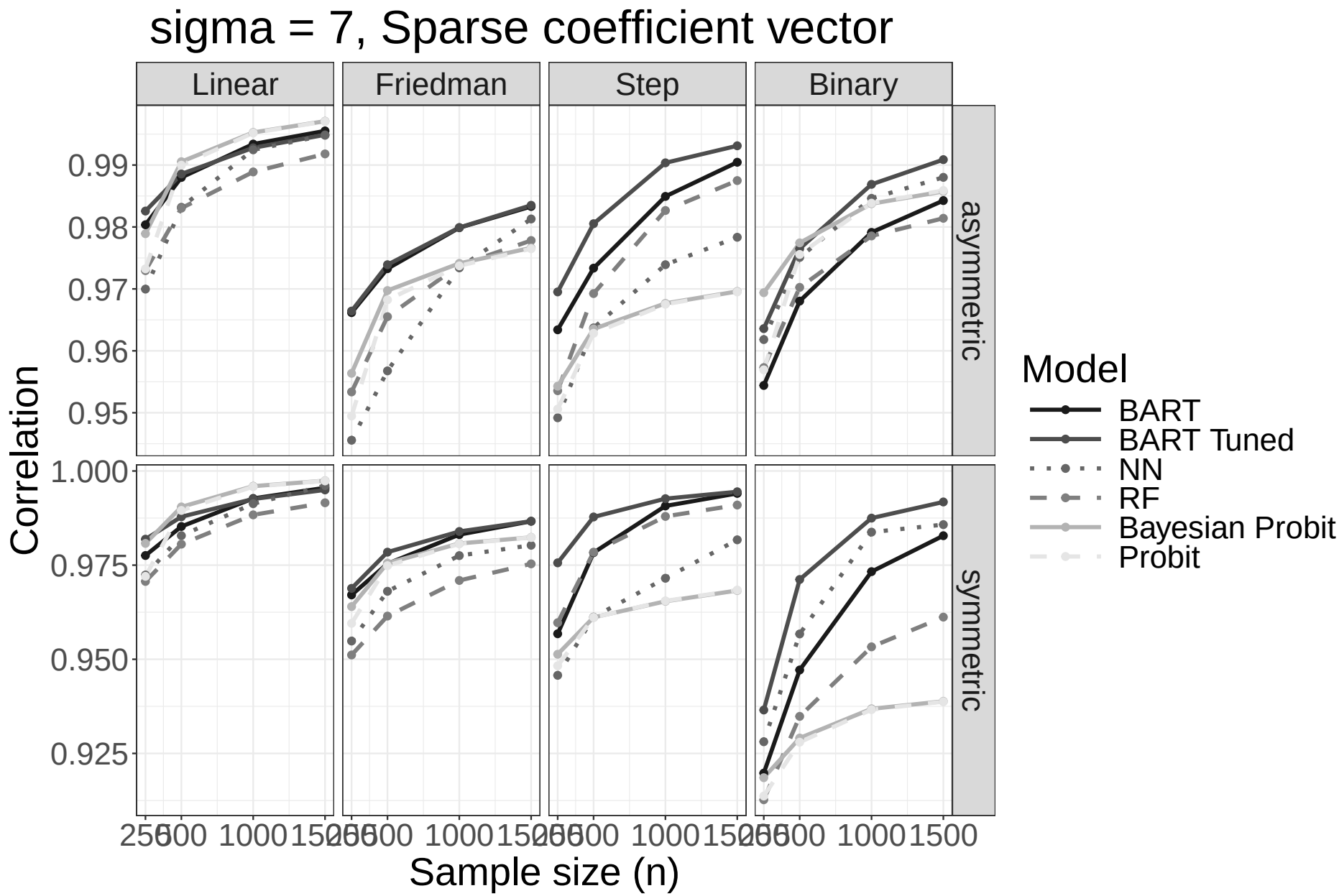
Simulation Diagnostics: Correlation, Relative RMSE, and Bias

Correlation plots



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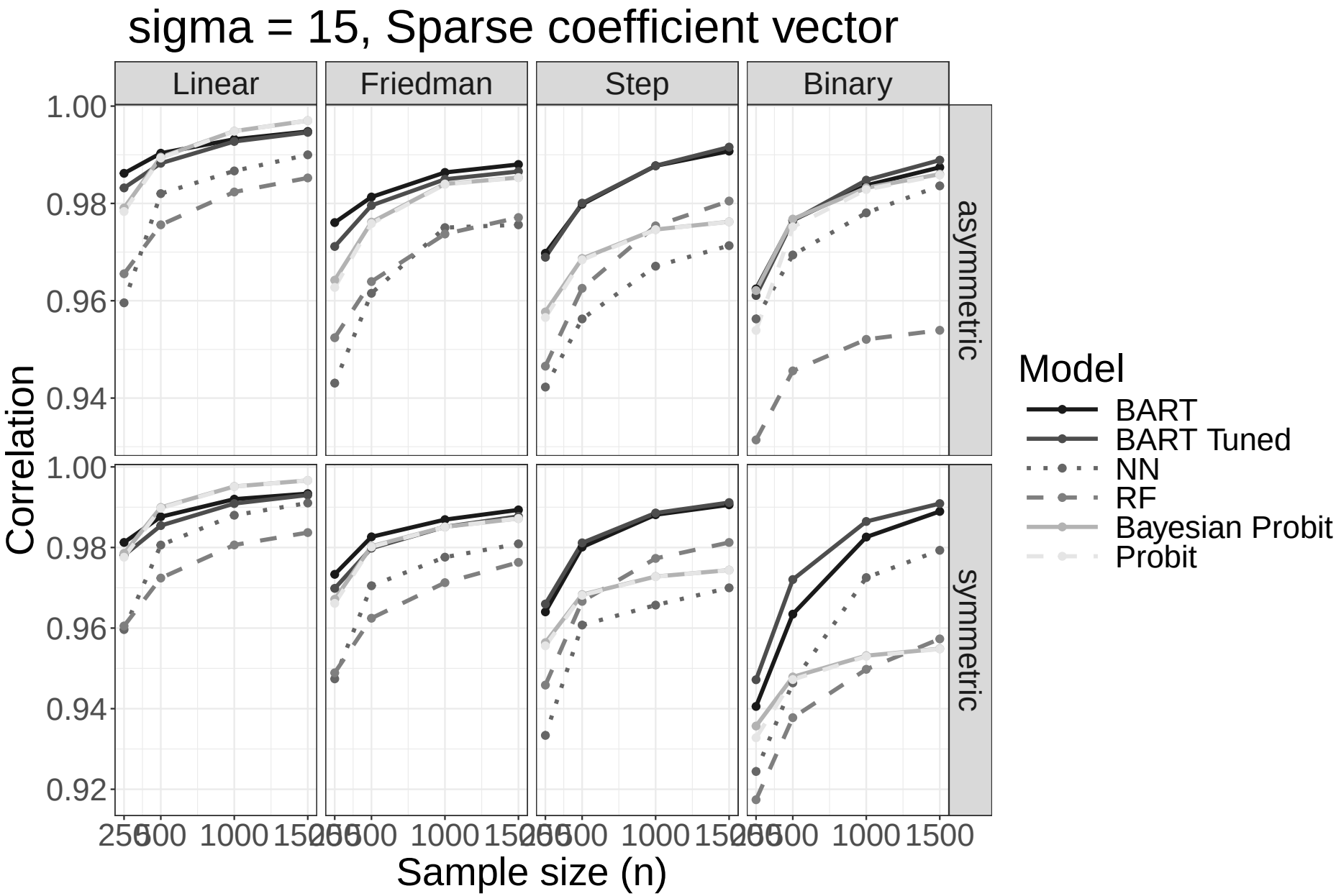


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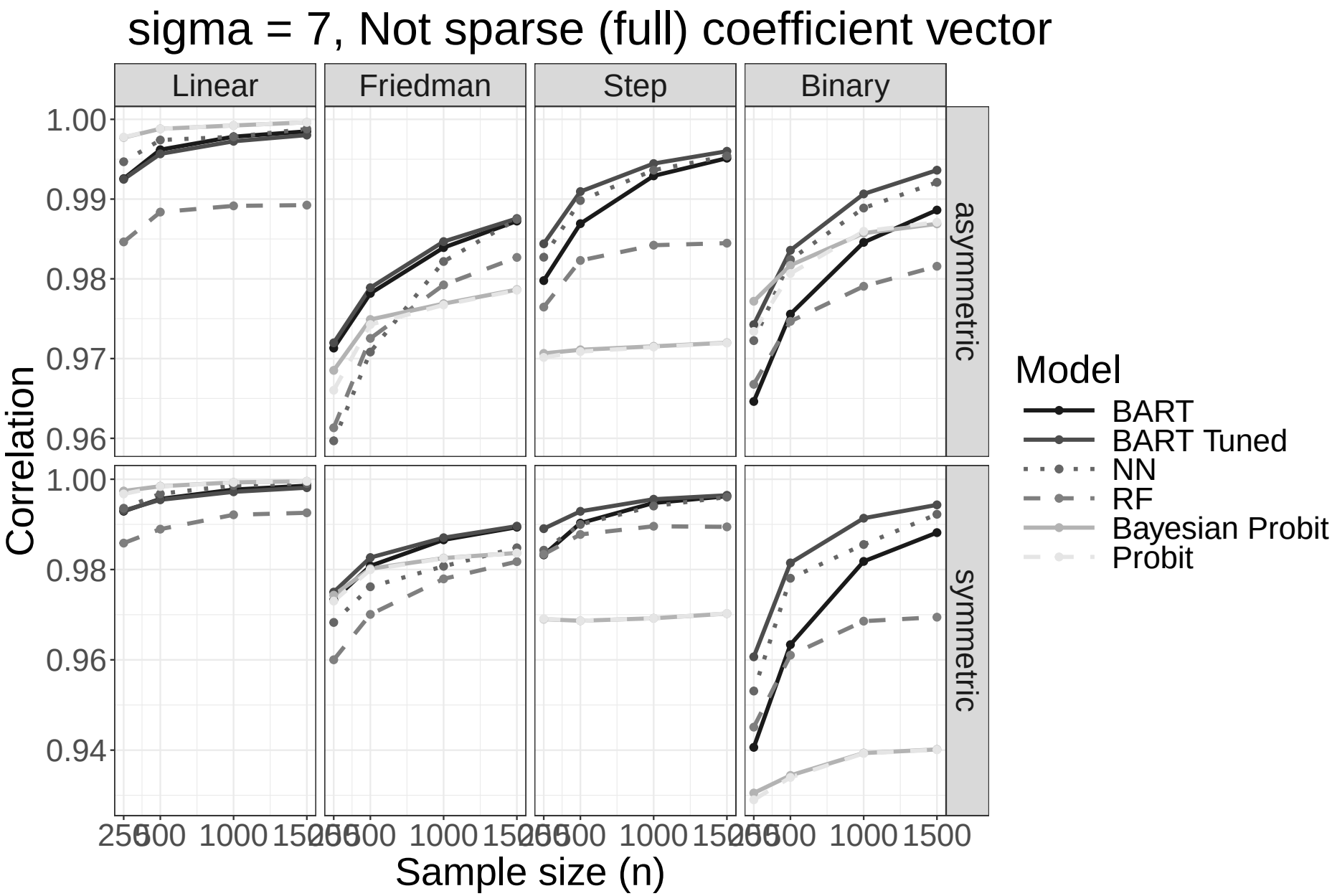
sigma = 7, Not sparse (full) coefficient vector
Linear
Friedman
Step
Binary
asymmetric
symmetric
Correlation
1.00
0.99
0.98
0.97
0.96
0.94
250
500
1000
1500
Sample size (n)
Model
BART
BART Tuned
NN
RF
Bayesian Probit
Probit

```
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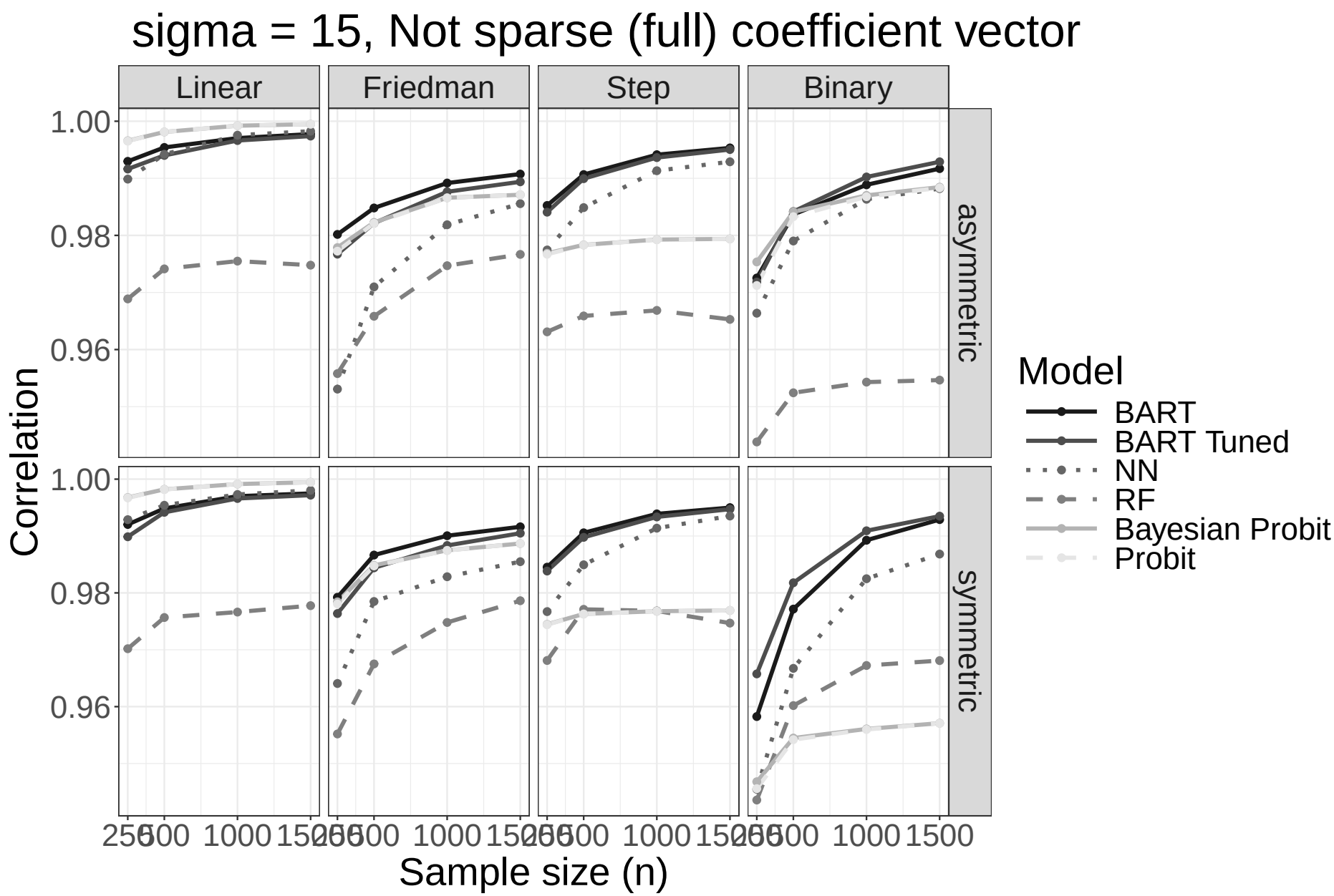
sigma = 15, Not sparse (full) coefficient vector
Linear
Friedman
Step
Binary
asymmetric
symmetric
Correlation
1.00
0.98
0.96
Sample size (n)
Model
BART
BART Tuned
NN
RF
Bayesian Probit
Probit

```
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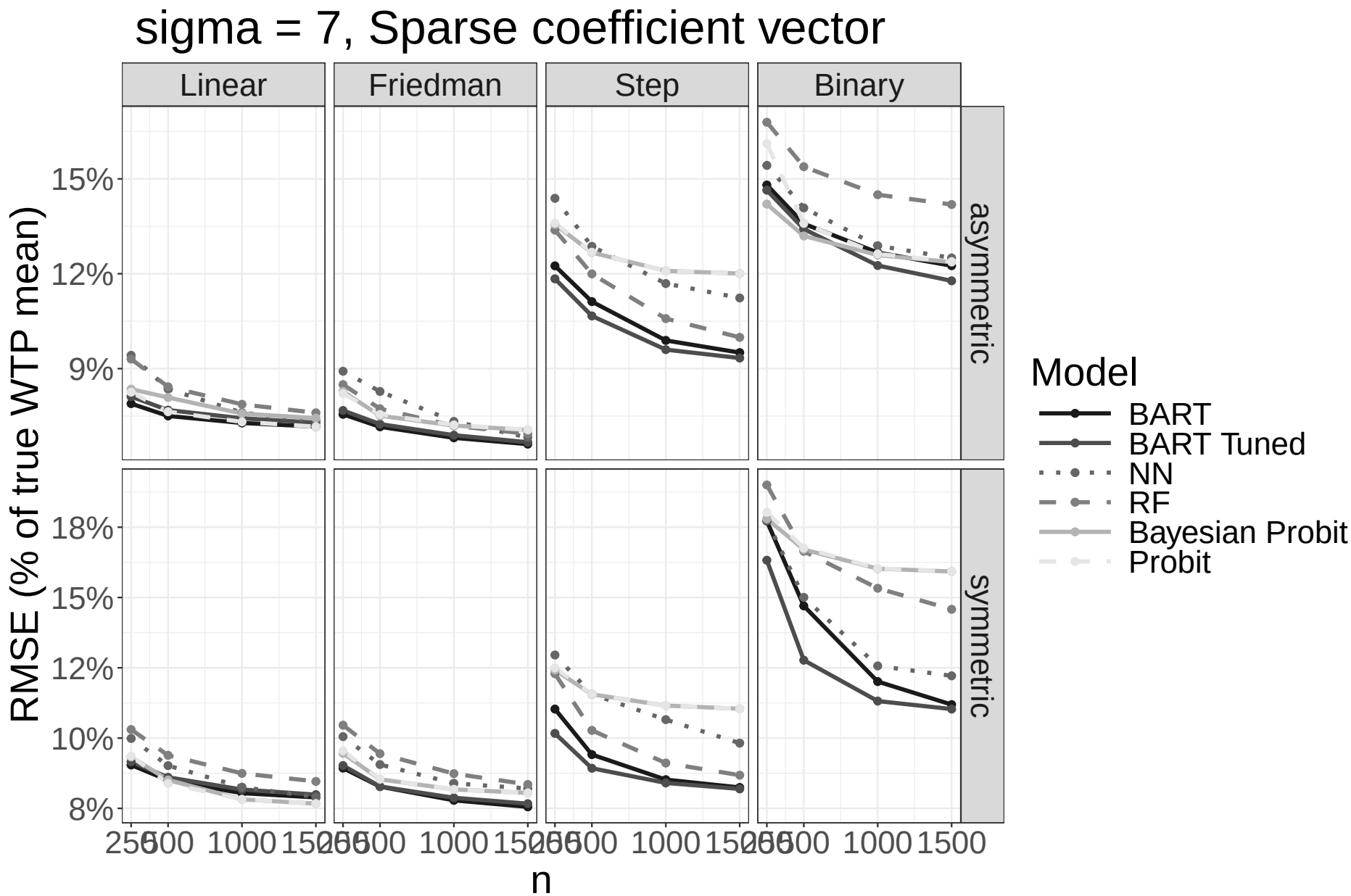
```

Relative RMSE plots (RMSE / mean(true)) as percent



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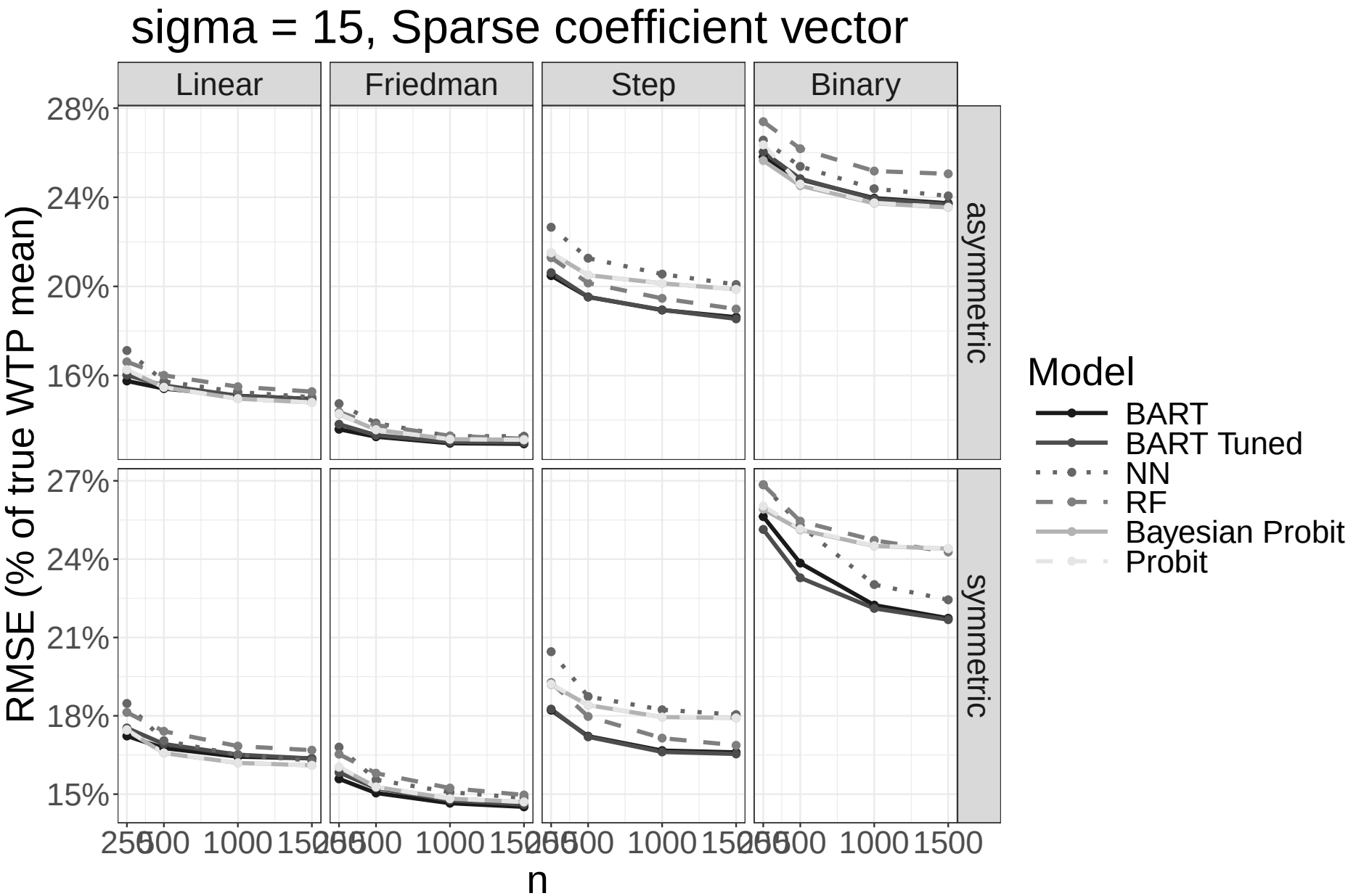


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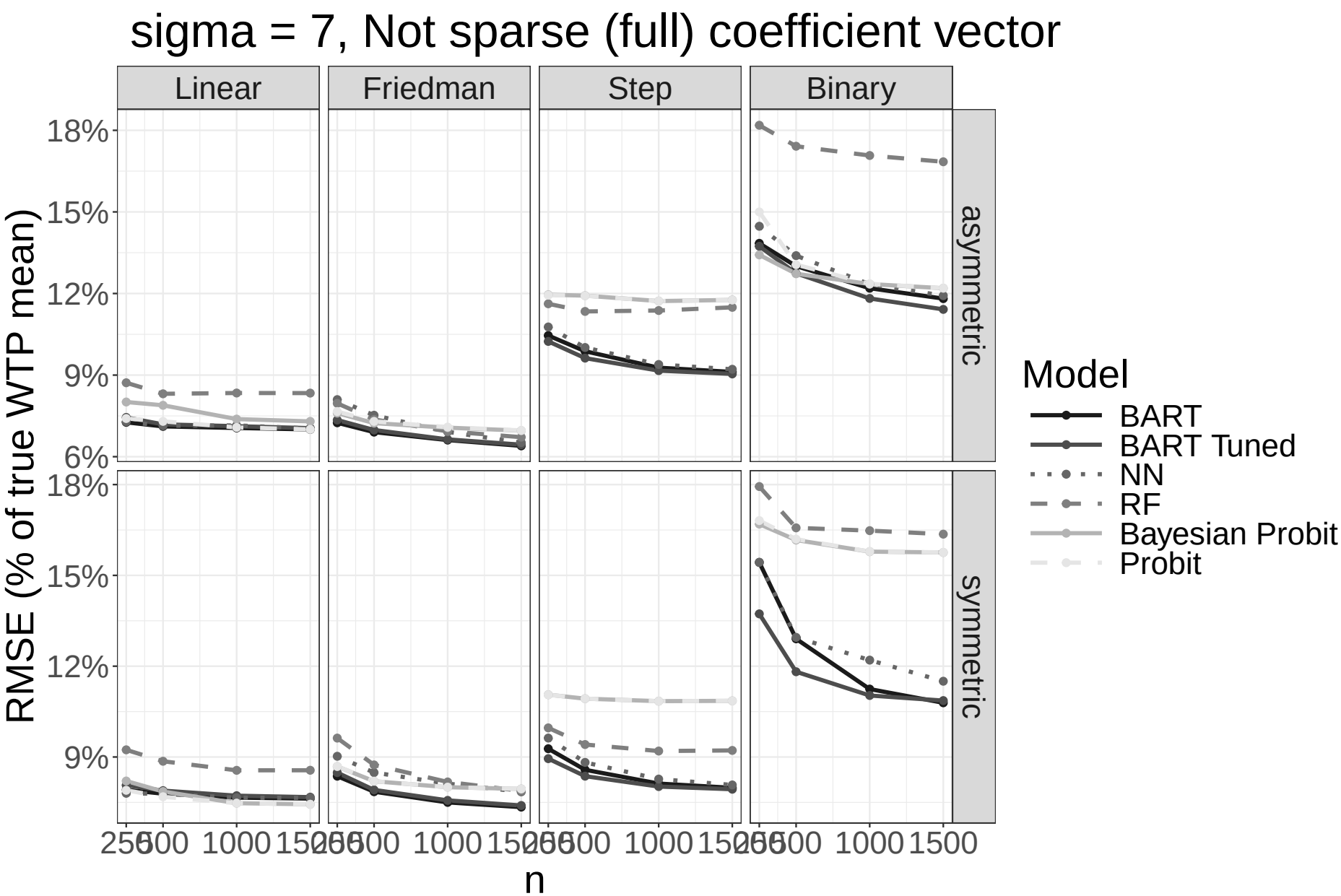
sigma = 7, Not sparse (full) coefficient vector
Linear
Friedman
Step
Binary
asymmetric
symmetric
RMSE (% of true WTP mean)
18%
15%
12%
9%
6%
n
Model
BART
BART Tuned
NN
RF
Bayesian Probit
Probit

```
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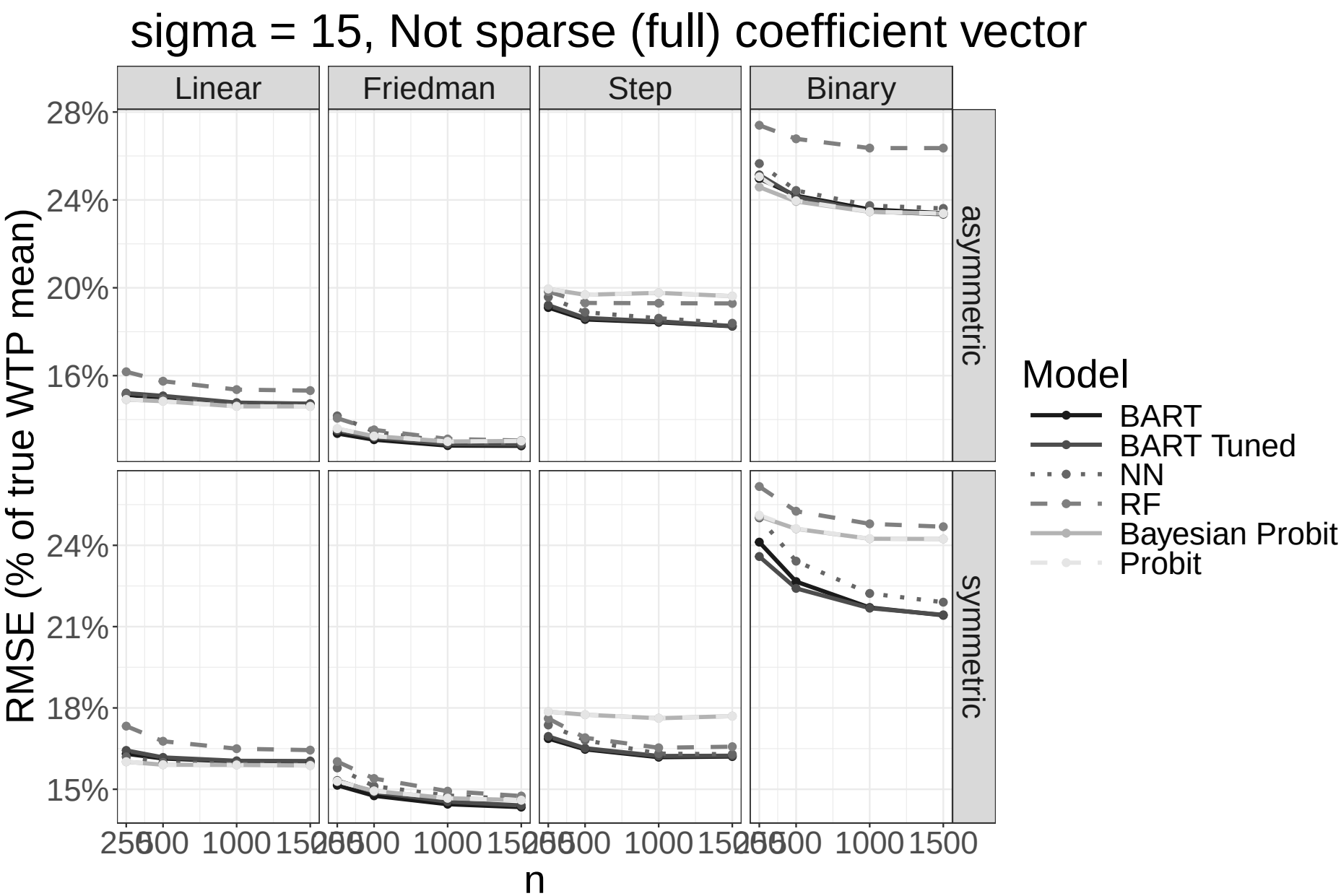
sigma = 15, Not sparse (full) coefficient vector
Linear
Friedman
Step
Binary
asymmetric
symmetric
RMSE (% of true WTP mean)
28%
24%
20%
16%
24%
21%
18%
15%
250
500
1000
1500
n
Model
BART
BART Tuned
NN
RF
Bayesian Probit
Probit

```
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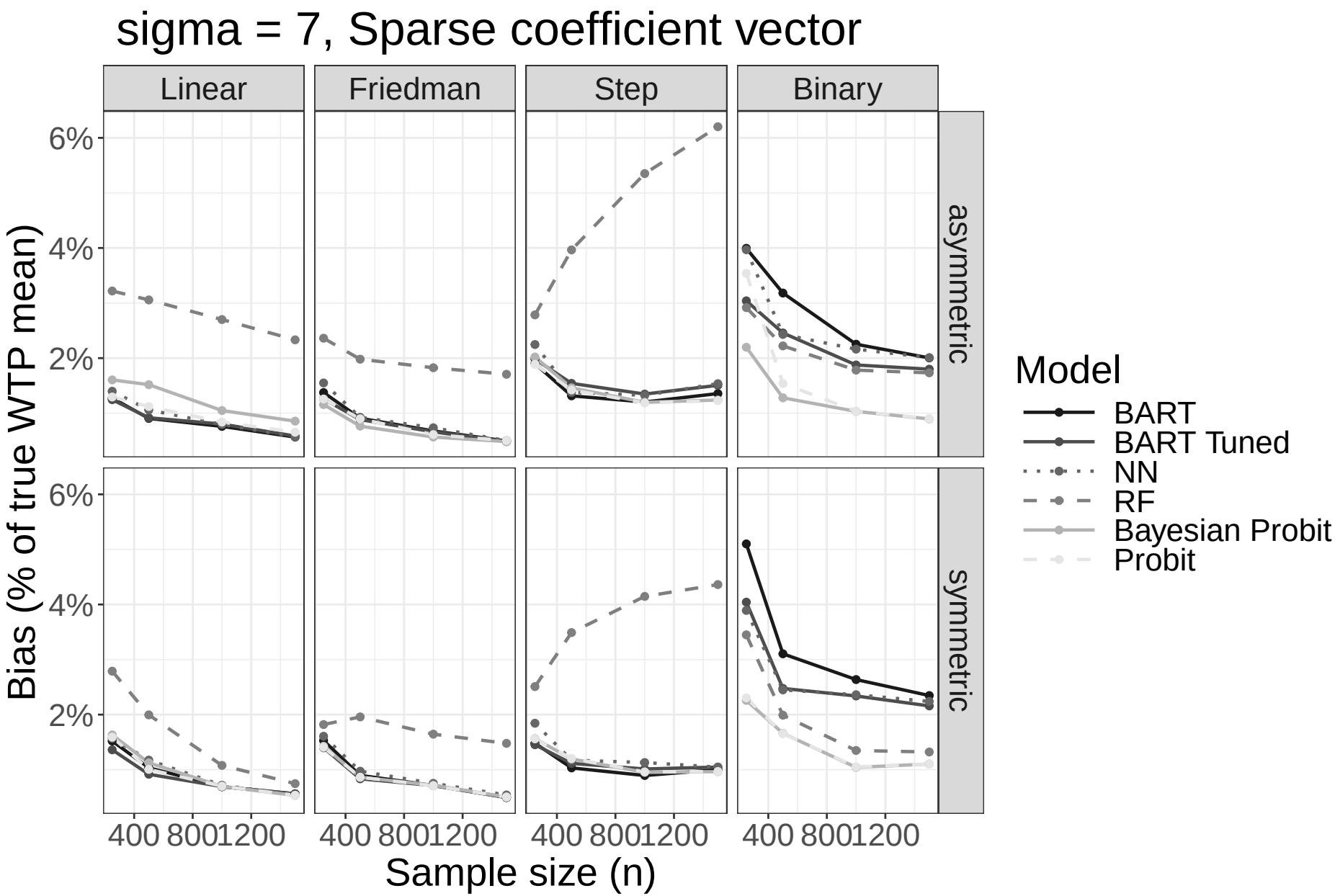
```

Bias plots


sigma = 7, Sparse coefficient vector
Linear
Friedman
Step
Binary
asymmetric
symmetric
6%
4%
2%
Bias (% of true WTP mean)
400 800 1200
Sample size (n)
Model
BART
BART Tuned
NN
RF
Bayesian Probit
Probit

```
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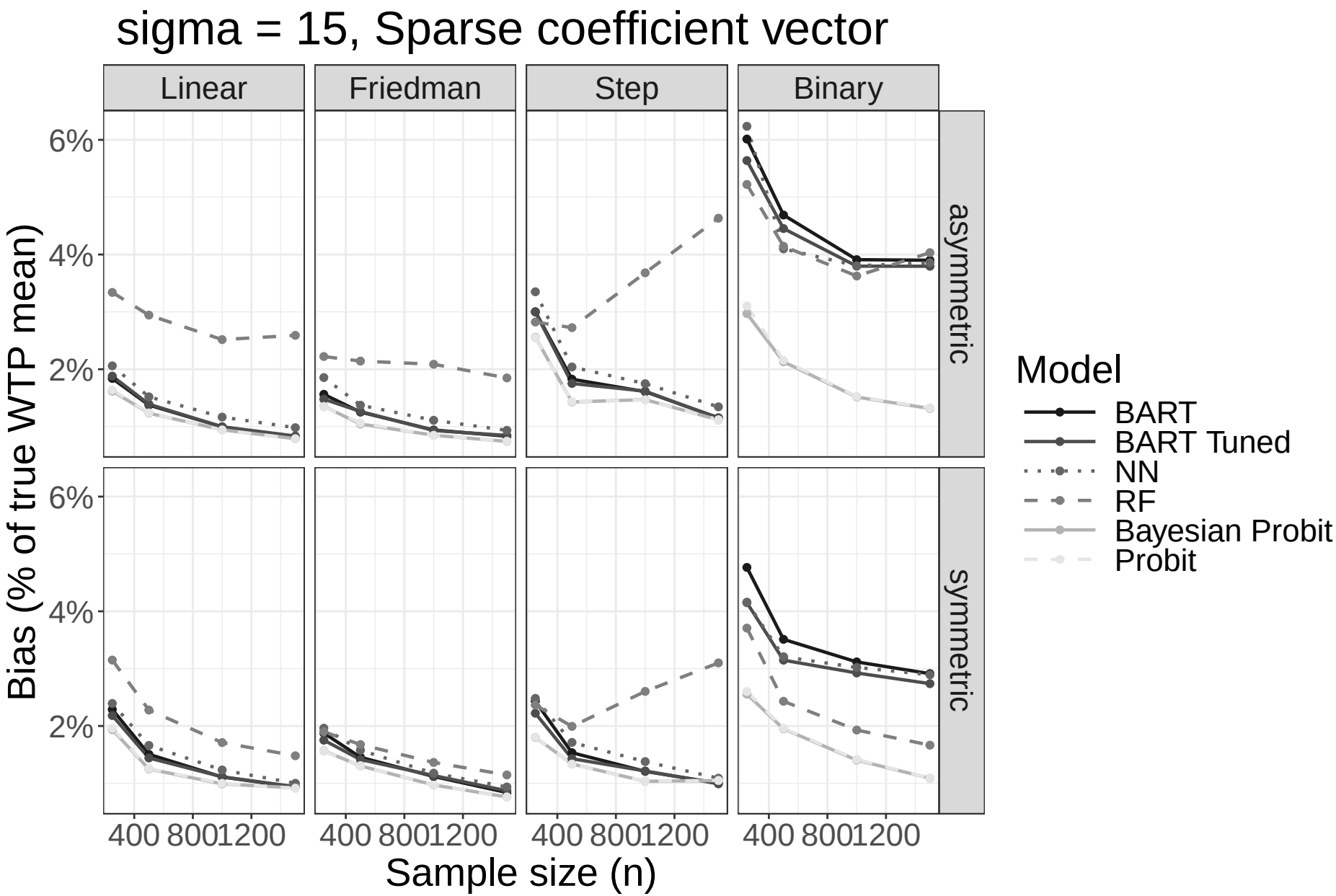


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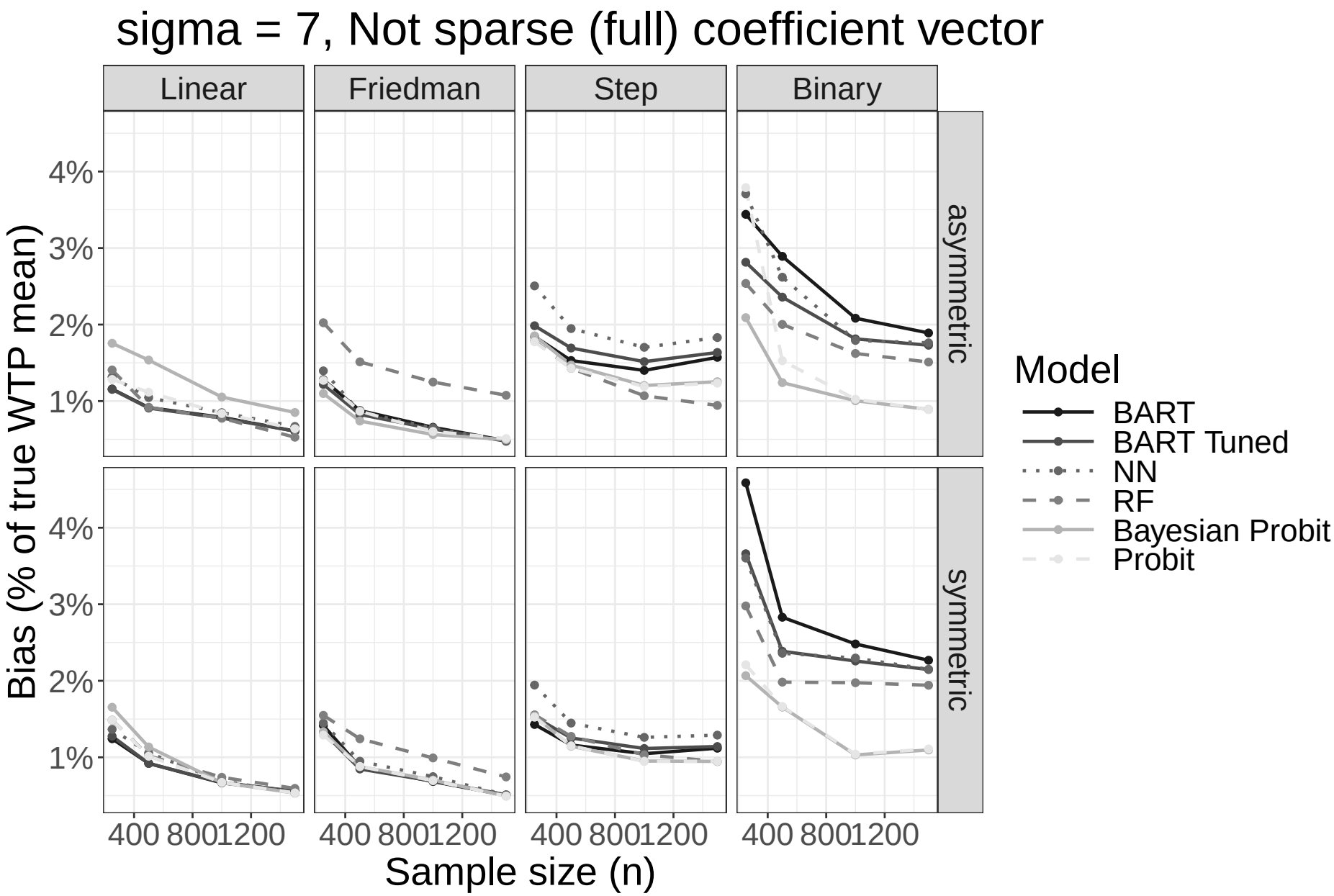
sigma = 7, Not sparse (full) coefficient vector
Linear
Friedman
Step
Binary
asymmetric
symmetric
Bias (% of true WTP mean)
4%
3%
2%
1%
400 800 1200
Sample size (n)
Model
BART
BART Tuned
NN
RF
Bayesian Probit
Probit

```
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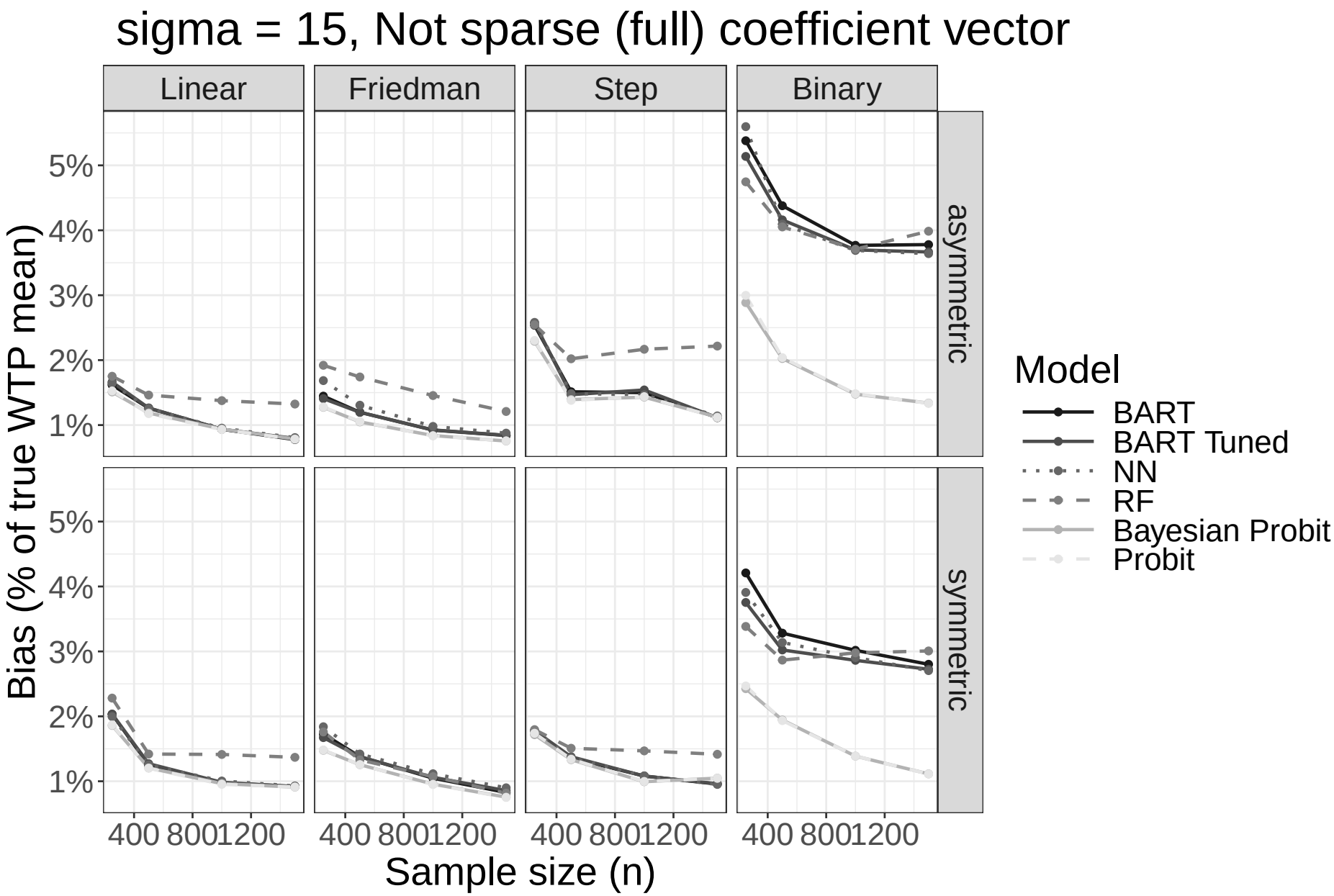
```


sigma = 15, Not sparse (full) coefficient vector
Linear
Friedman
Step
Binary
asymmetric
symmetric
Bias (% of true WTP mean)
5%
4%
3%
2%
1%
400 800 1200
Sample size (n)
Model
BART
BART Tuned
NN
RF
Bayesian Probit
Probit

```
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```